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$$f(x) = e^{\left(\frac{\sqrt[3]{\sin(x)}}{x}\right)}$$

$$f'(x) = e^{\left(\frac{\sqrt[3]{\sin(x)}}{x}\right)} \cdot \frac{d}{dx} \left(\frac{\sqrt[3]{\sin(x)}}{x}\right)$$

$$= e^{\left(\frac{\sqrt[3]{\sin(x)}}{x}\right)} \cdot \frac{\left(\left(x \cdot \frac{d}{dx} \left(\sqrt[3]{\sin(x)}\right)\right) - \sqrt[3]{\sin(x)}\right)}{x^2}$$

$$= e^{\left(\frac{\sqrt[3]{\sin(x)}}{x}\right)} \cdot \frac{\frac{x \cos(x)}{3\sqrt[3]{\sin^2(x)}} - \sqrt[3]{\sin(x)}}{x^2}$$

$$= e^{\left(\frac{\sqrt[3]{\sin(x)}}{x}\right)} \cdot \frac{\frac{x \cos(x) - 3 \sin(x)}{3\sqrt[3]{\sin^2(x)}}}{x^2}$$

$$= \frac{e^{\left(\frac{\sqrt[3]{\sin(x)}}{x}\right)} (x \cos(x) - 3 \sin(x))}{3x^2 \sqrt[3]{\sin^2(x)}}$$

References